III SEMESTER	L	T	P	C
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20AM3T02 : ARTIFICIAL INTELLIGENCE				

COURSE OUTCOMES:

Upon completion of this course, the students will be able to

CO1: Comprehend different types of problem-solving agents and its applications.

CO2: Solve problems using informed and uninformed search strategies.

CO3: Compare various Knowledge Representation Logic using scripts and frames.

CO4: Comprehend and analyze the different types of learning.

CO5: Identify the need of Production system and Planning states.

CO6: Use expert system tools to realize the concepts and components of expert system.

Unit – I:

PROBLEM SOLVING: Introduction to AI- Agents and Environments – Uninformed Search Strategies(Breadth first search, Depth first search) - Informed Search Strategies(Greedy best-first search- A* search, Hill climbing) - Problem Formulation-Constraint Satisfaction Problem.

Unit – II:

NOWLEDGE REPRESENTATION: Introduction to Game Playing-Alpha Beta Pruning-Knowledge Representation using First order logic-Knowledge Engineering in First Order Logic-Proportional vs First Order Logic-Resolution-Structured representation of Knowledge Using Scripts and Frames.

Unit – III:

INFERENCE AND LEARNING: Inference- Forward and Backward Chaining-Unification-Uncertainty-Inference in Bayesian Network –Learning from Observations-Forms of Learning-Inductive Learning-Neural Network-Learning Decision trees-Reinforcement Learning-Case Study-Learning examples with python.

Unit – IV:

PRODUCTION SYSTEM AND PLANNING: Introduction to Production system-control strategies-Rete Algorithm-Planning-STRIPS-Planning with state space search-Partial Order Planning-Planning Graphs-Planning & acting in the real world.

Unit – V:

EXPERT SYSTEM: Expert System- Architecture and Roles of Expert System-Typical Expert System-MYCIN-XOON-DART-Case Study-Construction of simple reflex agent with sensor and actuator using Arduino.

TEXT BOOKS:

1. Stuart Russell, Peter Norvig, “Artificial Intelligence – A Modern Approach”, 3rd Edition, Pearson Education / Prentice Hall of India, 2010.
2. Joseph C. Giarratano, Gary D. Riley, “Expert Systems: Principles and Programming”, 4th Edition, 2015.

REFERENCE BOOKS:

1. Nils J. Nilsson, “Artificial Intelligence: A new Synthesis”, Harcourt Asia Pvt. Ltd., 2000.
2. Janakiraman, K. Sarukesi, “Foundations of Artificial Intelligence and Expert Systems”, Macmillan Series in Computer Science, 2000.
3. W. Patterson, “Introduction to Artificial Intelligence and Expert Systems”, Prentice Hall of India, 2003.
4. Prateek Joshi, “Artificial Intelligence with Python”, Packt Publishing, 2017.

E BOOKS:

1. https://www.pdfdrive.net/artificial-intelligence-a-modern-approach-3rd-edition-e32618455.html

MOOC:

1. https://www.coursera.org/learn/introduction-to-ai